

Switching to TITE-CRM after nine patients at L1

Nine patients have been treated at 5×5 Gy and one of them recorded an acute DLT that could not be excluded as treatment-related. This note asks whether it is sound to amend to a TITE-CRM at this point, and whether the design still arrives at a defensible dose given that DLT.

SIMULATIONS 3000 PER CELL PATIENTS REMAINING 21 ACUTE TARGET 0.20

CONFIGURATION NO EWOC, NO BURN-IN

1 The answer, before the detail

Amending is sound, and the DLT at L1 is no longer the obstacle it was in July. The reason is arithmetic rather than statistical: one event in nine patients is a very different observation from one in six.

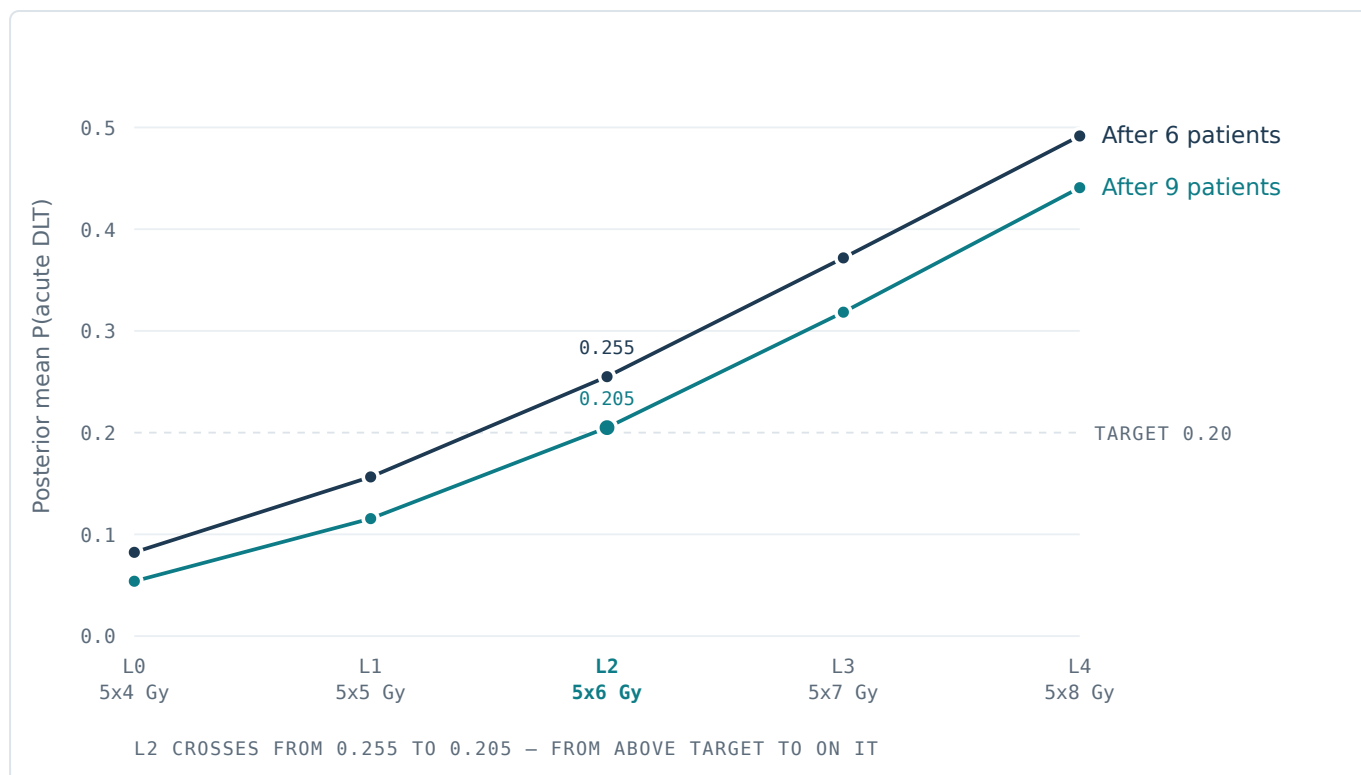
The observed rate falls from 0.167 to 0.111, comfortably below the 0.20 target, and eight DLT-free patients now pull the whole fitted curve down with it. That is enough to move the model's recommendation off L1.

RECOMMENDATION

Amend, and use the closest-to-target assignment rule. The model's first recommendation is L2 — the same dose the 6+3 rules already permit — so the amendment does not itself force a change of dose. Over the remaining 21 patients the model-based design ends at L1 or lower in 0–11% of trials against 16–50% for 6+3, at a cost of roughly 0.5–0.9 additional acute DLTs.

2 What the model says today

This is not a simulation. It is the posterior fitted to the nine patients actually treated, with the prior skeleton from the July protocol and complete acute follow-up.



The three additional patients move L2 onto the target. In July the posterior put L2 at 0.255, above the 0.20 target, and the design recommended staying at L1. With nine patients L2 sits at 0.205 — effectively on target — and the recommendation becomes L2.

POSTERIOR ACUTE TOXICITY AND THE RESULTING DOSE, BY STATE OF THE L1 DATA								
DATA AT L1	L0	L1	L2	L3	L4	CLOSEST RULE	HIGHEST-BELOW RULE	
6 patients, 1 DLT (July state)	0.082	0.157	0.255	0.372	0.492	L1	L1	
9 patients, 1 DLT (current state)	0.054	0.115	0.205	0.318	0.441	L2	L1	
9 patients, 0 DLT (counterfactual)	0.007	0.021	0.047	0.094	0.163	L4	L4	
9 patients, 2 DLT (counterfactual)	0.112	0.204	0.315	0.439	0.557	L1	L0	
12 patients, 1 DLT (if L1 grew further)	0.039	0.091	0.173	0.283	0.405	L2	L2	

The recommendation is L2 whether the contested DLT is counted at full weight, at half, or not at all.

That last point is worth stating plainly to the team. In July the attribution of the single contested event decided the dose. It no longer does: with nine patients on the level, counting the DLT at attribution 1.0, 0.5 or 0.0 all yield L2 under the closest-to-target rule. The adjudication discussion remains relevant to the trial record, but it has stopped being load-bearing for the next dose decision.

3 What the model does next

A model-based design is easier to accept if the committee can see its behaviour rather than be asked to trust it. The table below is exhaustive for the next cohort: every possible outcome of three patients at L2, and the dose that follows. It is computed directly from the posterior, not sampled.

WHAT THE MODEL DOES AFTER THE FIRST COHORT OF THREE AT L2					
ACUTE DLTS AT L2	P(DLT) L1	P(DLT) L2	P(DLT) L3	NEXT DOSE	MOVE
0 of 3	0.081	0.159	0.266	L2	stay at L2
1 of 3	0.136	0.235	0.355	L2	stay at L2
2 of 3	0.193	0.305	0.430	L1	down to L1
3 of 3	0.254	0.373	0.496	L1	down to L1

The design moves one level at a time and never skips. One DLT in three at L2 is not enough to send it back down; two is. Escalation to L3 requires six DLT-free patients at L2. Compared with the 6+3 rules currently in force, which stop and de-escalate on two DLTs in six, this is the more conservative response to an isolated event and the less conservative response to an accumulation of clean data — which is the trade the amendment is really making.

4 Two definitions of the right answer

Before the simulation results, one caveat that changes how they read.

The running protocol defines the MTD as the highest dose with acute toxicity at or below 0.20. A CRM with the closest-to-target rule is built to find the dose *nearest* 0.20. In three of the five scenarios these are the same dose. In **Acute high** (L2 = 0.15, L3 = 0.22) and **Acute steep** (L2 = 0.08, L3 = 0.24) they are not: the nearest dose is L3, the highest dose under target is L2.

Scoring the design against the protocol definition alone would mark it wrong for meeting its own objective, so both are reported throughout. This is not a presentational choice — it is a decision the amendment has to make explicitly, because the assignment rule and the MTD definition must be the same rule.

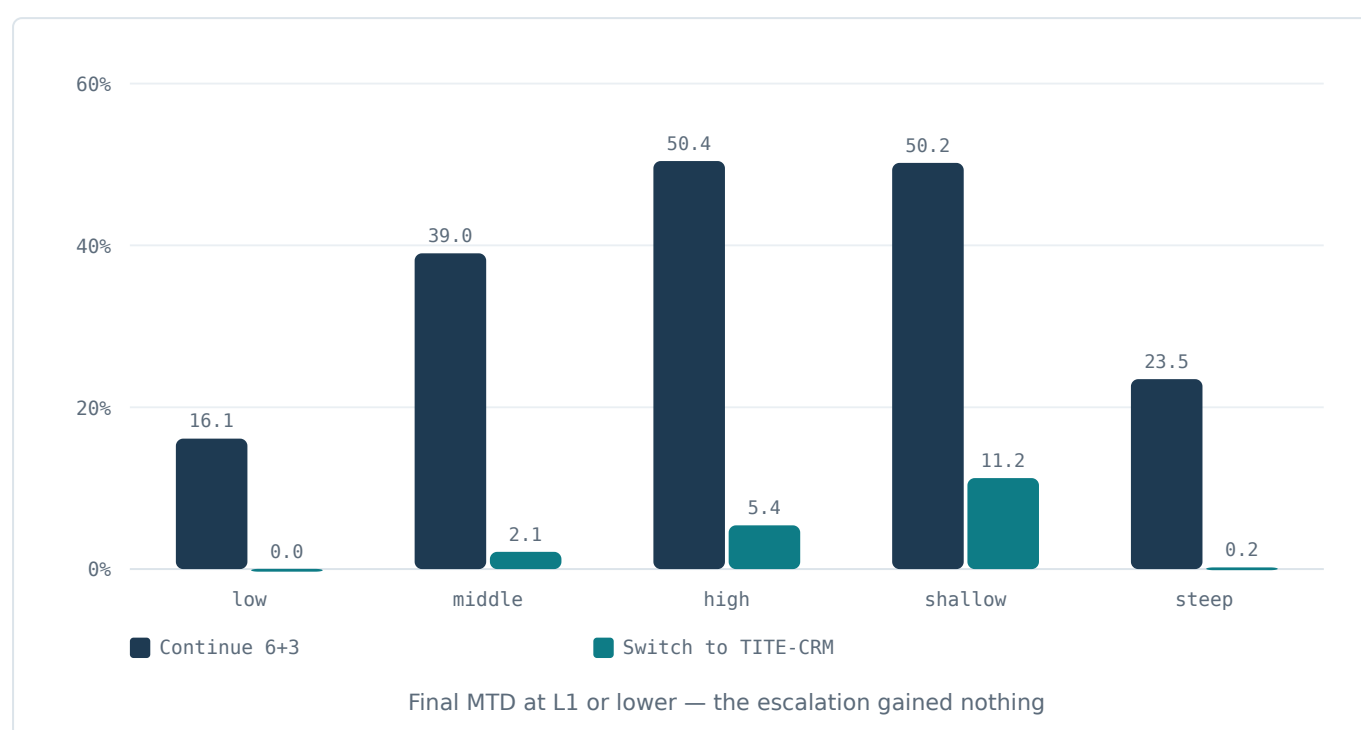
CORRECT DOSE SELECTED (%), UNDER BOTH DEFINITIONS					
DESIGN · DEFINITION OF CORRECT	LOW	MIDDLE	HIGH	SHALLOW	STEEP
Continue 6+3 · Highest dose $\leq$ 0.20	0.0	7.5	45.7	5.5	71.6
Continue 6+3 · Closest to 0.20	0.0	7.5	3.9	5.5	5.0
Switch to TITE-CRM · Highest dose $\leq$ 0.20	18.5	45.2	60.6	35.1	44.6
Switch to TITE-CRM · Closest to 0.20	18.5	45.2	32.3	35.1	53.3
TITE-CRM, R dose rule · Highest dose $\leq$ 0.20	0.0	16.9	67.0	7.8	72.4
TITE-CRM, R dose rule · Closest to 0.20	0.0	16.9	9.5	7.8	23.6

Compare the two designs within a definition, not across them. Under the protocol definition the model-based design is ahead of 6+3 in four of five scenarios and behind in Acute steep, where it is deliberately choosing L3. Under its own definition it is ahead everywhere, by wide margins.

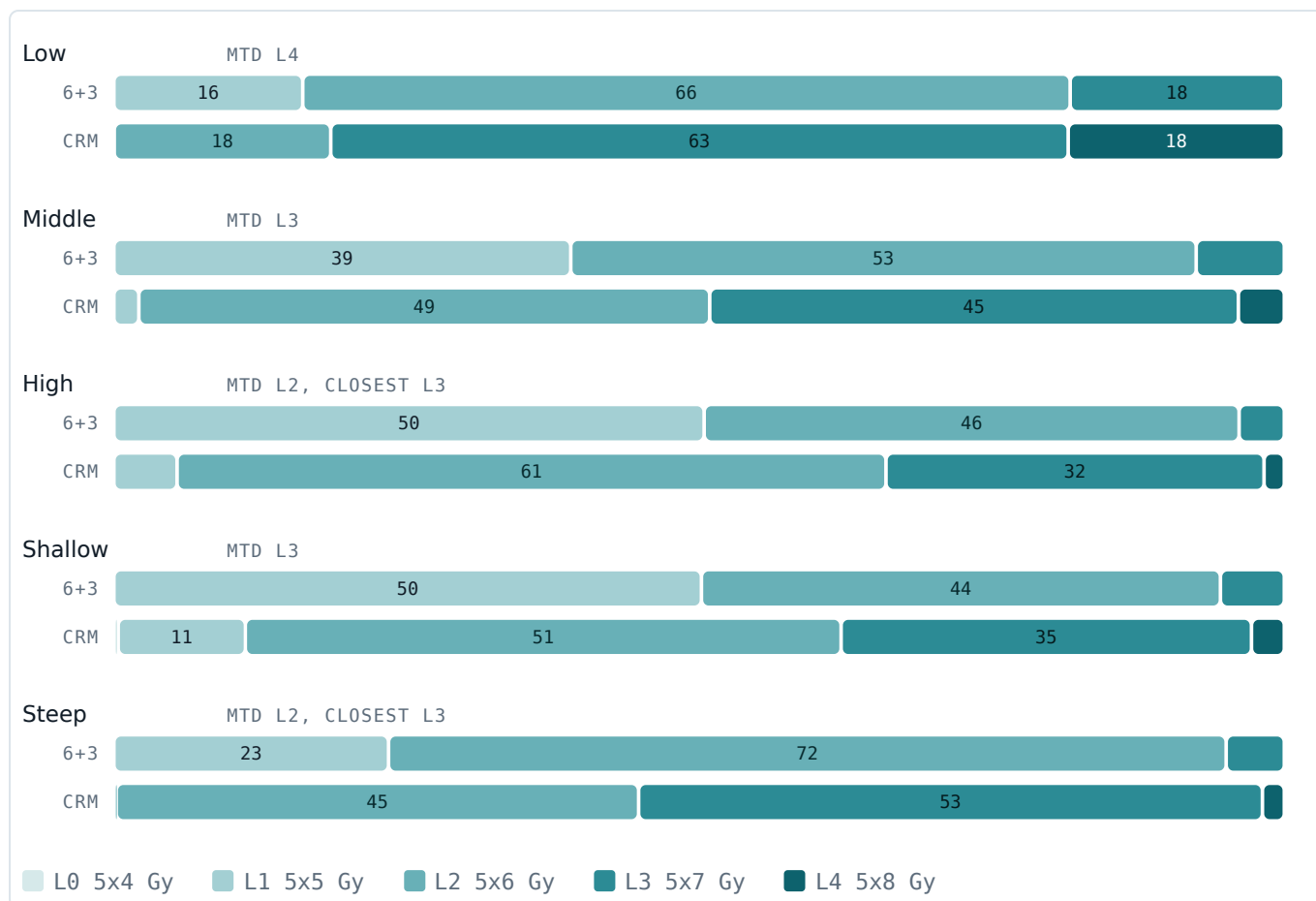
The bottom pair is the R implementation's rule, included here because it is the live alternative. It should be read fairly: under the protocol definition it is the *better* estimator in exactly the two scenarios where the definitions diverge (Acute high 67.0% against 60.6%, Acute steep 72.4% against 44.6%), which is what a rule that never crosses the target should do. What it gives up is everything else: Acute low 0.0%, Acute middle 16.9%, Acute shallow 7.8%.

5 The risk that has changed hands

In July the concern was that a model-based design would lock onto L1 and never escalate. With 21 patients left and nine required per level, that risk now belongs to the design currently in the protocol.



Probability the trial ends at L1 or lower. 6+3 falls back to the dose already given in 16.1–50.4% of trials; the model-based design in 0.0–11.2%. In Acute high, where the true toxicity at L2 is only 0.15, the chance of seeing two or more DLTs among nine patients is about 40% — enough to stop 6+3 and send it back to L1 half the time.



Where each design finishes. Each bar is the distribution of the final selected dose over 3000 simulated continuations. The 6+3 rows are concentrated on L1 and L2; the model-based rows shift a substantial share to L3, and in Acute low reach L4 at all.

6 What it costs

The model-based design treats patients at higher doses, so it produces more toxicity. The question is how much, and whether it ever recommends something unsafe.

WHAT EACH DESIGN COSTS IN TOXICITY					
DESIGN · MEASURE	LOW	MIDDLE	HIGH	SHALLOW	STEEP
Continue 6+3 · Mean acute DLTs in the 21 new patients	1.2	2.1	2.4	2.8	2.0
Continue 6+3 · True acute toxicity of the selected dose	0.061	0.093	0.102	0.142	0.074
Continue 6+3 · Selected dose has true toxicity > 0.30 (%)	0.0	0.0	0.0	0.0	0.0
Switch to TITE-CRM · Mean acute DLTs in the 21 new patients	1.7	2.8	3.3	3.2	2.9
Switch to TITE-CRM · True acute toxicity of the selected dose	0.102	0.147	0.170	0.161	0.170
Switch to TITE-CRM · Selected dose has true toxicity > 0.30 (%)	0.0	0.0	0.0	0.0	1.9

The additional burden is roughly half to one acute DLT across the 21 remaining patients. The dose finally recommended carries a true acute toxicity of 0.10-0.17 on average, against a 0.20 target. A dose with true toxicity above 0.30 is selected in at most 1.9% of trials, and only in Acute steep, where L3 sits at 0.24 and the fourth level at 0.34. There is no scenario in which the amended design runs away from the data.

One asymmetry to note in the comparison: 6+3 stops early when it stops for toxicity, so it uses 18.7–20.3 of the 21 remaining patients on average, while the model-based design always uses all 21. A difference of one to two patients — not material to the comparison, but it is there.

7 The one decision still open

The assignment rule matters more than anything else in this note, and the two candidate rules disagree about what to do today.

Under the highest-dose-below-target rule used in the R implementation, the recommendation given the current data is **L1, not L2**. Amending under that rule would mean stepping back from the dose the 6+3 rules already permit. That rule also never reaches L4 in any of the five scenarios, and in Acute shallow it ends at L1 or lower in 45.2% of trials.

This is the same divergence documented in the earlier simulator-comparison note, now with a concrete consequence attached. Whichever rule is adopted, the amendment must state it together with the matching definition of the MTD.

TO BE SPECIFIED IN THE AMENDMENT

The assignment rule (closest to target, or highest below target); the matching MTD definition; the one-level step limit; the requirement that no dose above the highest yet tried may be assigned; how the contested L1 DLT enters the likelihood; and whether the subacute endpoint constrains dose assignment. In the configuration analysed here the subacute endpoint is modelled and reported but does not drive assignment, which reflects the current code and should be confirmed rather than assumed.

8 Limits of this analysis

- The nine treated patients are represented as nine observations at L1 with one acute DLT and complete follow-up. Their surgery status is drawn from the 0.80 rate used elsewhere in the simulator rather than taken from the record.
- No subacute DLTs are assumed among them, per the trial team.
- The 6+3 comparator restarts its level evaluation at L2 with 21 patients. It carries no memory of the L1 cohort beyond the fact that L1 was passed, which is what the protocol rules actually use.
- Scenario probabilities are the five acute scenarios from the July report, unchanged. They are assumptions, not estimates.
- The differences between this simulator and the R implementation described in the accompanying comparison note are unresolved. The numbers here are internally consistent; they are not yet reconciled with Sama's.

Simulations: 3000 per design and scenario, seed 20260910. Acute target 0.20, subacute target 0.33, prior scale 1.0, one-level step limit, no EWOC, no burn-in, final MTD restricted to tried doses.

Reproduced by `switch_decision_analysis.py` and `switch_decision_tree.py`; outputs in `switch_decision/`.

This document contains aggregated simulation output only. It contains no patient data.